

Acceleration of an Antarctic outlet glacier driven by surface meltwater input to the base

Received: 25 October 2025

Accepted: 20 April 2026

Published online: 06 May 2026

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Despite its importance and being commonly accepted in other regions, meltwater influence on the Antarctic ice sheet dynamics continues to be debated and questioned. To investigate the possible impact of surface melt on subglacial water pressure and ice dynamics, we performed hot-water drilling ~1 km upglacier from the grounding line of Langhovde Glacier in East Antarctica. Borehole measurements revealed that the subglacial water pressure exceeded 90% of the ice overburden and the pressure elevated during periods of intensive melting and rain. Coinciding with these events, ice speed increased by 10–20% and the surface rose by ~0.1 m. Subglacial water was freshwater, but hydraulically connected to the sub-shelf cavity, where sessile animals were distributed in a thin seawater layer. Our in-situ measurements confirm meltwater-driven acceleration of grounded ice in Antarctica. Since meltwater is ubiquitous along the Antarctic coast, its impact on outlet glacier dynamics should be considered when projecting Antarctic ice sheet evolution.

The ice flow speed of Antarctic outlet glaciers plays a crucial role in the mass balance of the ice sheet, as it controls the discharge of ice into the ocean. Fast-flowing glaciers have shown accelerating trends over the last few decades, affecting recent ice loss along the Antarctic coast^{1–3}. The fast flow of Antarctic glaciers and ice streams is attributed to basal ice motion⁴, which occurs through ice sliding over the base or subglacial sediment deformation^{5,6}. These subglacial processes occur at melting conditions where water pressure has a strong influence^{7–9}. Borehole observations in mountain glaciers have shown ice acceleration during periods of elevated water pressure^{8,10–12}, and similar observations have been reported in Greenland¹³. In Antarctica, glacier speed-up associated with subglacial water pressure variations have been reported only from an ice cap on an island in the Peninsula region¹⁴. For outlet glaciers of the Antarctic ice sheet, no evidence has been reported for short-term ice speed variations associated with changing subglacial water pressure.

Meltwater is broadly distributed along Antarctic ice sheet margins^{15,16}. While interior regions of the Antarctic ice sheet remain too cold for substantial surface melt, many coastal and ice-shelf zones do produce significant meltwater during the summer season. Recent data indicate that meltwater features, including surface lakes, ponds, and slush, are widespread around the ice-sheet margin

and are increasing in some regions^{17,18}. The potential influence of surface meltwater on subglacial environments is crucial for ice sheet evolution as it can amplify ice discharge into the ocean¹⁹. Nevertheless, the penetration of surface meltwater to the glacier base has never been confirmed in Antarctica, and the influence of surface melt on glacier dynamics has been a topic of controversy. Satellite data indicates seasonal variations in ice speed on the Antarctic Peninsula^{20–22}. Because this seasonal speed pattern was in phase with glacier surface melting, enhanced basal motion caused by meltwater drainage to the base was suggested as the possible driver. However, this mechanism has been debated because changes in the buttressing force are also likely under the influence of the ice front position, ice shelf thickness, and sea ice conditions^{23–25}. Moreover, precise grounding-line locations are required to accurately attribute ice speed changes to subglacial processes²⁶.

Despite the debate surrounding the drainage of surface water to the base, the existence of subglacial hydrological systems in Antarctica is commonly accepted^{27,28}. Satellite observations and numerical investigations indicate water movement between subglacial lakes²⁹, as well as subglacial discharge at the grounding line^{30–32}. Basal hydrology and hydraulic conditions near the grounding line are essential not only for ice dynamics but also for ice shelf basal melting^{33,34} and subglacial

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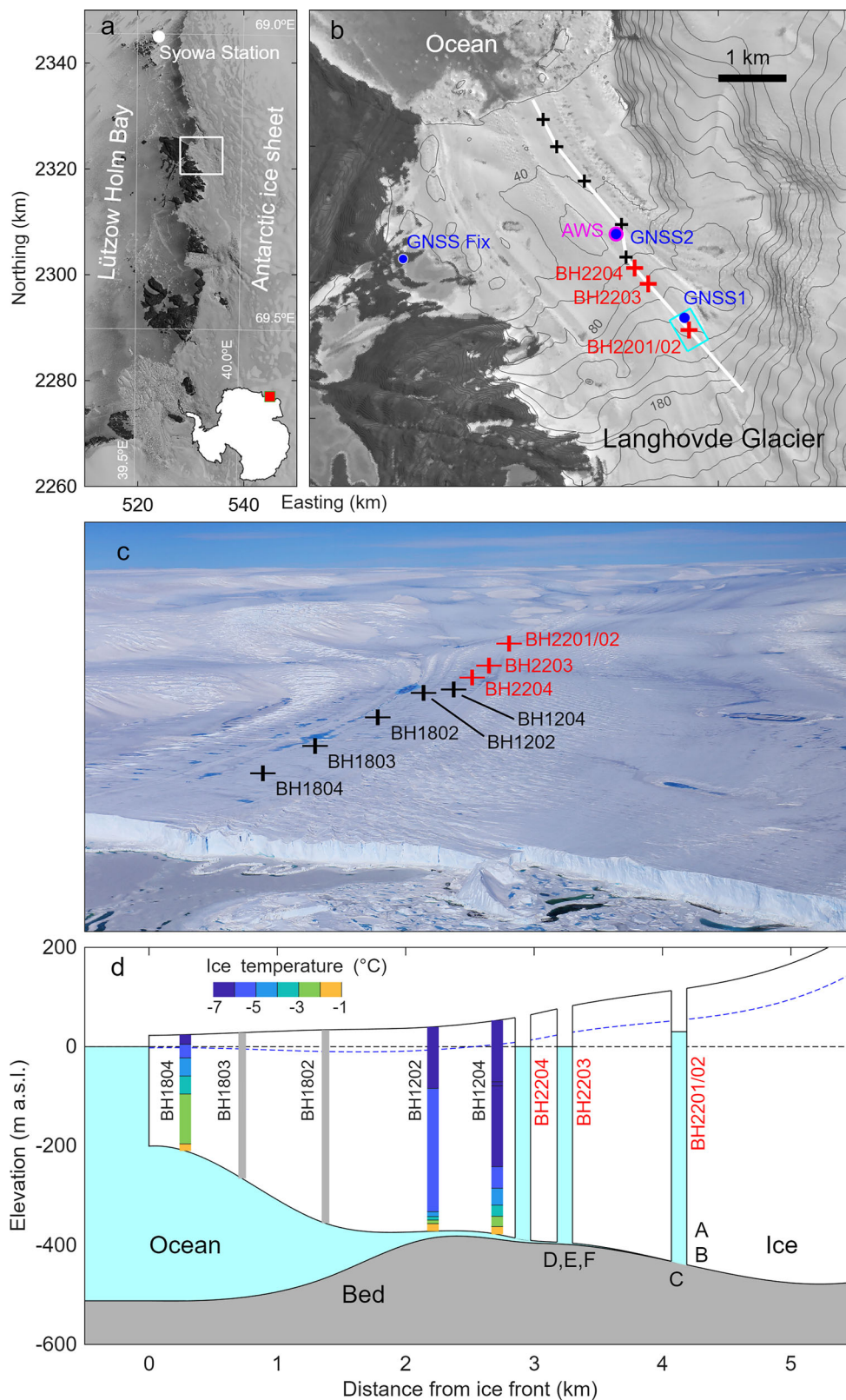


Fig. 1 | Satellite image, aerial photograph, and vertical cross-section of Langhovde Glacier. **a** The location of the study site in Antarctica. White box shows the area covered by **(b)**. **b** The study area of the glacier with locations of boreholes (+) drilled in this study (red) and previous studies (black), GNSS (●), and automatic weather station (AWS) (○). The white line connecting the boreholes was used to generate the cross-section shown in **(d)**. The box in light blue indicates the area covered by the drone images shown in Fig. 4a–c. The contours are surface elevation in m a.s.l. with intervals of 20 m based on

the Pleiades digital elevation model on 17 December 2021. The background of **(a, b)** is the Landsat 8 image on 18 December 2021 distributed by USGS. **c** Oblique photograph of the glacier taken from a helicopter on 14 December 2021. **d** Vertical cross-section of Langhovde Glacier along the line indicated in **(b)**. The black and blue dashed curves are ocean surface and flotation levels (Methods). The color scale indicates the ice temperature measured in boreholes BH1202, BH1204, and BH1804. A–F indicate the approximate locations of the borehole video camera images in Fig. 2a–f, respectively.

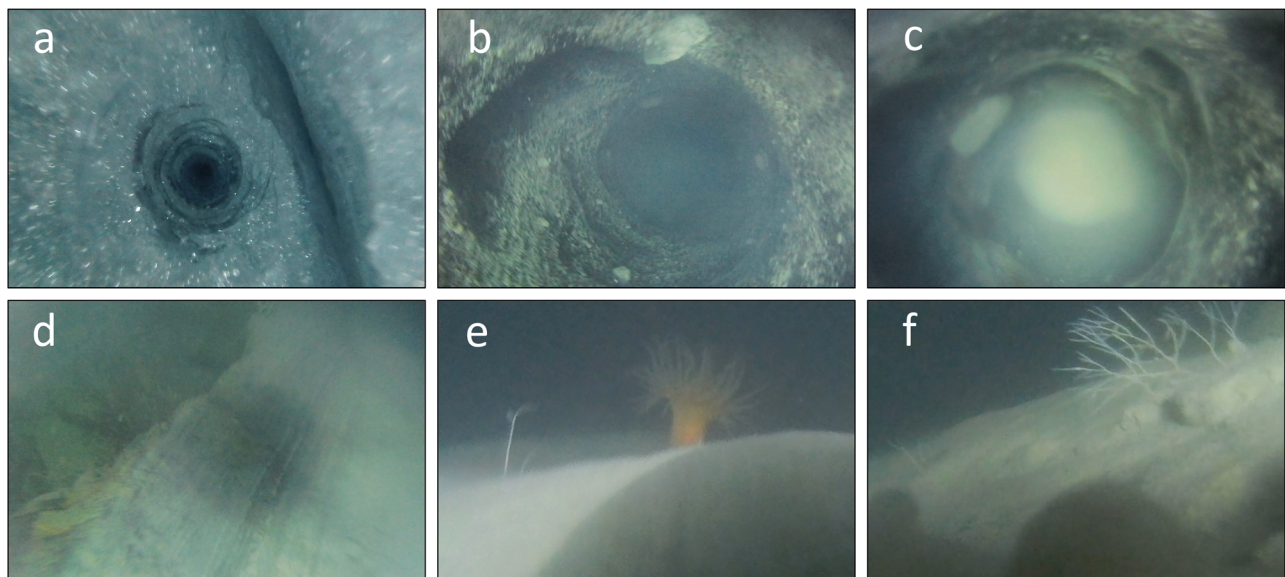


Fig. 2 | Borehole and sub-shelf images taken at A–F in Fig. 1d. a Ice crack 41 m above the bed, **b** sediment and rock inclusions ~2 m above the bed and **c** the bottom of the borehole in BH2201. The borehole diameter is ~100–200 mm. **d** Boulder,

e sea anemone, and **f** stalked sponges found on the seafloor beneath BH2203. Image widths are ~200–300 mm.

biogeochemical processes³⁵. Nevertheless, direct observations beneath grounded ice in Antarctica are sparse.

Results

Hot-water drilling and borehole inspection

From December 2021 to February 2022, we conducted hot-water drilling, borehole and glacier surface measurements on Langhovde Glacier, a 3-km-wide outlet glacier located on Lützow-Holm Bay, East Antarctica (Fig. 1). Results from previous drilling in 2011/2012 and 2017/2018 show the lower 3 km of the glacier forms an ice shelf with an ice thickness of ~430 m in the grounding zone^{36,37} (Fig. 1d). The drilling sites in the 2021/2022 season were situated in the upper reaches of the boreholes from the previous campaigns (Fig. 1b, c), with a focus on the basal conditions under the grounded ice.

The first borehole was drilled 4 km from the glacier front, and drilling reached the bed at a depth of 552 m from the glacier surface (440 m below sea level) (BH2201 in Fig. 1b–d). During drilling, water in the borehole drained at ~40 m above the bed, indicating a hydrological connection of the borehole presumably to a basal crevasse. The second borehole (BH2202) was drilled ~5 m from the first hole, and the drill reached the bed at a depth of 552.5 m. Water in the second hole drained during the drilling ~90 m from the bed. Two other boreholes were drilled at 3.2 and 3.0 km from the glacier front, which confirmed the presence of a shallow seawater layer under the ice (BH2203 and BH2204 in Fig. 1b–d). Thus, the grounding line was situated between BH2201/02 and BH2203, ~1 km upglacier from the point where the flotation level exceeds sea level (Fig. 1d), i.e., the transition from floating to grounded ice based on the hydrostatic equilibrium assumption. Ice and subglacial seawater cavity thicknesses were 474 and 3 m at BH2203 and 450 and 6 m at BH2204.

Video inspection of borehole BH2201 confirmed the existence of an englacial crevasse at the depth of water drainage during the drilling (Fig. 2a and Supplementary Movie 1). Ice within ~20 m from the base included sediment, which ranged in size from clay to gravel (Fig. 2b and Supplementary Movie 2), although the sediment concentration appears to be smaller than that observed near the bed of Ice Stream B in West Antarctica³⁸. The turbidity of the borehole water was relatively low, so the glacier base was visible (Fig. 2c and Supplementary Movie 2). According to a CTD (conductivity, temperature, and depth)

profiler cast, the salinity of the borehole water was below the detection limit (<0.01 PSU). The video camera lowered into borehole BH2203 captured the sub-shelf environment and fauna in the vicinity of the grounding line. A sea anemone (Actiniaria) and demosponges were observed on a boulder situated below the borehole (Fig. 2d–f and Supplementary Movie 3).

Ice speed and subglacial pressure variations

Ice flow speed showed significant temporal variations during the field measurement from December 2021 to February 2022. The observations were made by global navigation satellite system (GNSS) receivers installed near BH2201 and BH2202 (GNSS1) and 2 km downglacier from the boreholes below the grounding line (GNSS2) (Fig. 1b). Following a period of good weather with daily maximum temperatures of about 5 °C, ice speed increased and peaked on 24 December 2021 (Period I in Fig. 3a, c). Intensive ice melting was observed on the glacier during the relatively warm conditions, which lasted about five days. The peak speed at GNSS2 was approximately 20% faster than the background speed, and a similar or even greater acceleration was observed at GNSS1.

The second speed-up occurred on 2–6 January 2022 (Period II in Fig. 3a). The glacier was covered with low clouds, and the daily mean temperature was consistently above freezing from 2 to 5 January. Approximately 30 mm of rainfall was recorded on 2 January (Fig. 3c). Liquid precipitation is unusual in this region, i.e., the last recorded rain at Syowa Station was in December 2017. The glacier ice had been covered by a ~50 mm thick snow layer before the rain event (Fig. 4a), which was then soaked by rain and meltwater. On the morning of 3 January, surface snow was saturated entirely (Fig. 4b), and meltwater began draining into a crevasse ~200 m downglacier from boreholes BH2201 and BH2202 (Fig. 4d, e). Water-saturated snow gradually disappeared, and the glacier surface became drier 10 days after the event (Fig. 4c).

The speed-up events were associated with vertical glacier surface motion. During the first event from 21 to 25 December, GNSS1 showed an upward displacement of 120 mm (Fig. 3a). A smaller uplift (~10 mm from 3 to 5 January) was observed during the second event. Importantly, no noticeable vertical displacement was observed at GNSS2, where ice was afloat.

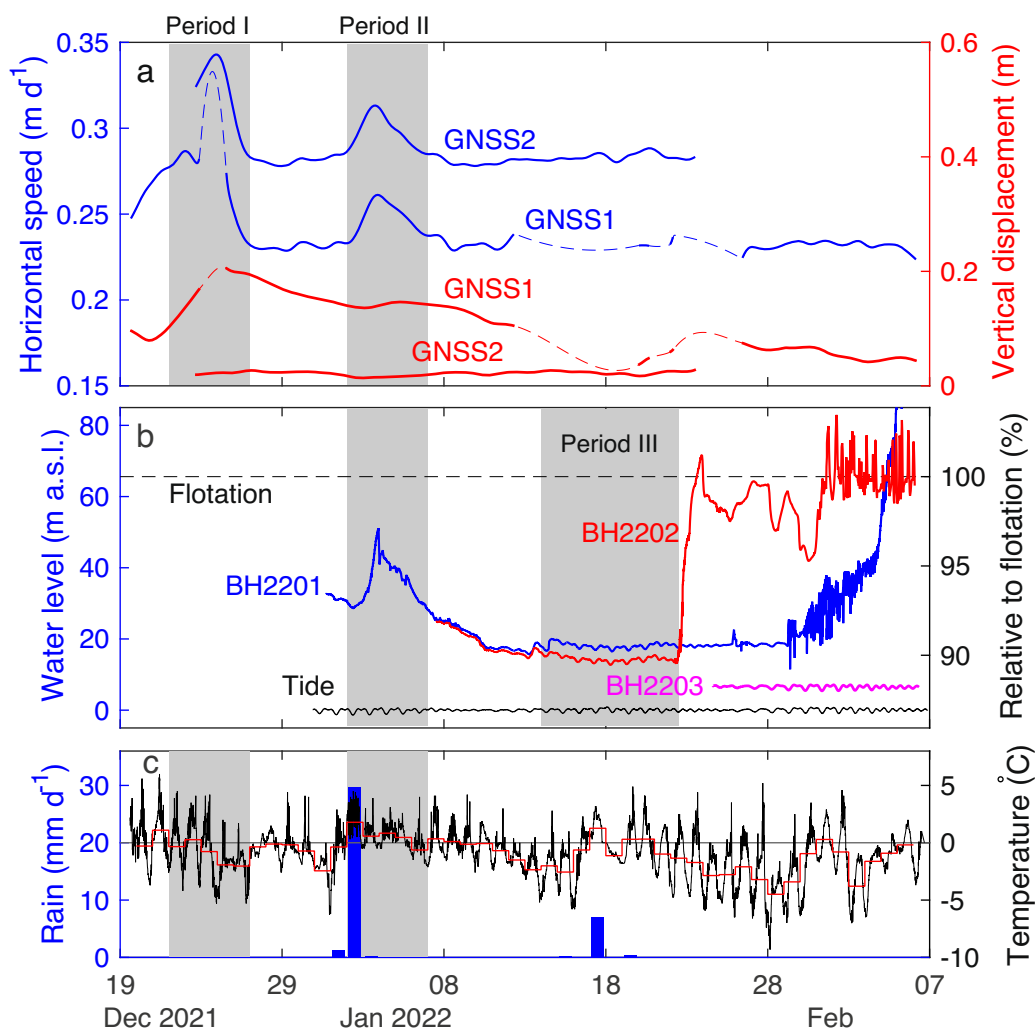


Fig. 3 | GNSS, borehole water pressure, and automatic weather station data. **a** Ice speed (blue) and vertical displacement (red) at GNSS1 and GNSS2. The dashed lines are estimates from the spline interpolation of the GNSS coordinates before and after the data gaps. **b** Water levels in boreholes BH2201 (blue), BH2202 (red), and BH2203 (magenta). Despite its connection to the sub-shelf cavity, the water level at BH2203 was -6 m above sea level because the borehole was filled with

freshwater. The black solid line indicates the ocean tide measured at Syowa Station. The dashed line is the flotation water level at BH2201/02. **c** Hourly (black), daily mean air temperature (red), and daily liquid precipitation (blue). Grey bands indicate the periods of speed-up events (Period I and Period II) and semidiurnal pressure variations (Period III). The bands are given only on panels, which are of interest and referred to in the text.

The second speed-up event coincided with an increase in the subglacial water pressure (Fig. 3b). After the installation of the pressure sensor at the bottom of borehole BH2201 on 31 December, the borehole water level was 29–33 m above sea level (a.s.l.) until 2 January. These water levels corresponded to 93–94% of the flotation level, i.e., the subglacial water pressure was equivalent to 93–94% of the ice overburden pressure. On January 3, the water level rose progressively by 20 m and peaked at 51 m a.s.l. (97% of the flotation). The timing of the pressure peak broadly coincided with that of the ice speed peak (Fig. 3a, b). The water level dropped by ~10 m within an hour and gradually decreased ~30 m over the following 10 days.

Beginning 14 January, the water level in borehole BH2201 exhibited semidiurnal variations with typical amplitudes smaller than 1 m (Period III in Fig. 3b). High-frequency oscillations in the borehole level began on 29 January, followed in early February by a rapid increase to above the flotation level. The second pressure sensor was installed in borehole BH2202 on 7 January. Water levels in the two boreholes were within 5 m agreement, and their variations continued to be very similar until 22 January, when the water level in BH2202 rapidly rose to the flotation level (Fig. 3b).

Discussion

Meltwater-driven glacier acceleration

The GNSS and borehole data, together with visual observations of glacier surface conditions, strongly suggest that surface meltwater penetrated the glacier to the base and enhanced basal sliding (Supplementary Fig. 1). Surface meltwater-driven glacier acceleration has been debated in the Antarctic Peninsula, but a link between ice dynamics and subglacial conditions has been difficult to prove using only satellite data^{20–25}. Our subglacial measurements and in-situ GNSS data confirm the influence of surface meltwater on subglacial pressure and its impact on glacier speed in Antarctica.

Temperature measurements in previously drilled boreholes indicated meltwater drained through cold ice (Fig. 1d and Supplementary Fig. 2). Although temperature (~–7 °C near the surface) was warmer than those reported in other Antarctic glaciers^{39,40}, ice was predominantly cold (i.e., the ice temperature was below the melting point), except for the glacier base, where melting conditions are expected due to relatively fast ice flow. Therefore, surface meltwater reached the bed through an englacial hydrological network formed in cold ice. Presumably, surface water flowed through a channel generated by hydrofracturing⁴¹, a mechanism proposed for

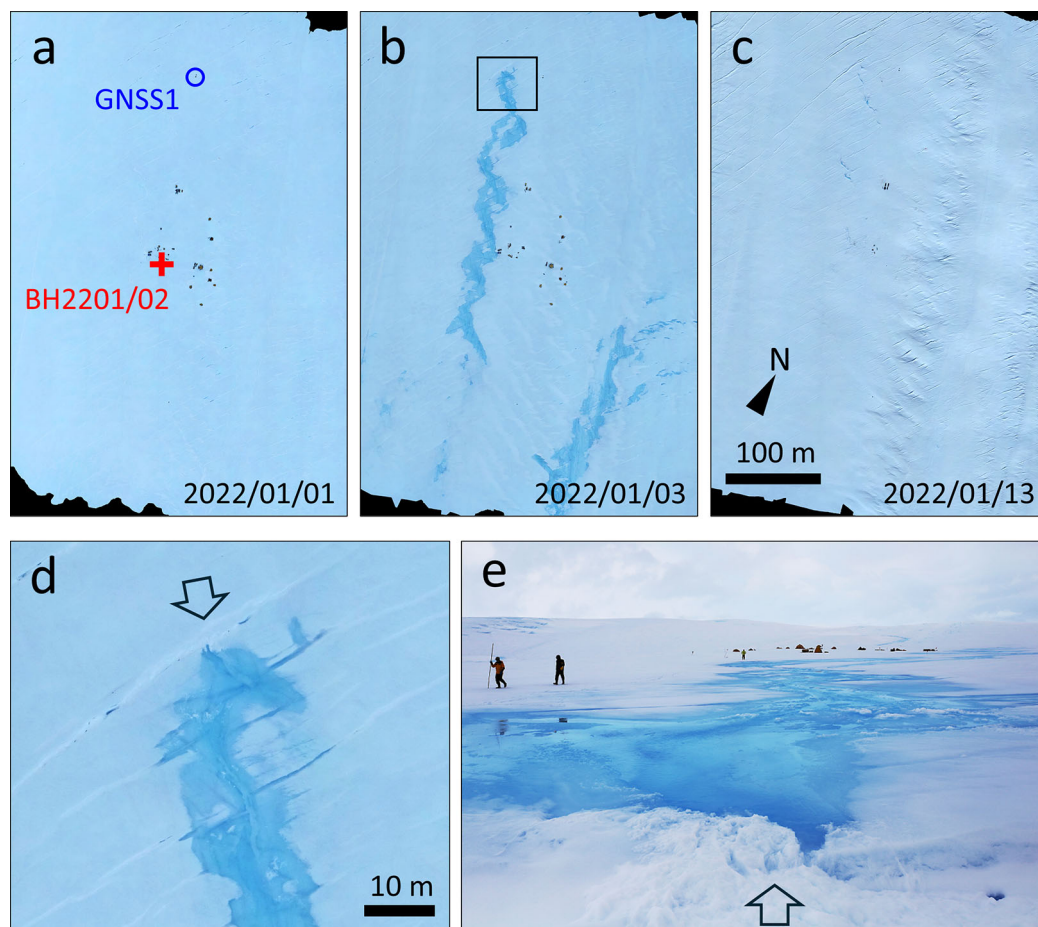


Fig. 4 | Glacier surface conditions around BH2201/02. **a–c** Drone images taken on 1, 3, and 13 January 2022 over the region indicated in Fig. 1b. The box in **b** indicates the area covered by **(d)**. **d** Enlarged image of the area shown in **(b)**. The arrow shows

the location of water drainage and the direction of the photograph shown in **(e)**. **e** Photograph taken on 3 January 2022, showing the water drainage into a crevasse.

supraglacial water drainage through Greenlandic glaciers^{42,43}. This process has also been suggested as the mechanism of supraglacial lake drainage observed on ice shelves in Antarctica^{44,45}. Hydrofracturing is implied by the observation of surface water draining into a crevasse during the increase in subglacial pressure and ice acceleration (Fig. 4e). It is also supported by the fracture observed in borehole BH2201 (Fig. 2a). Two of boreholes (BH2201 and BH2202) drained during drilling, thus we assume that basal crevasses were densely distributed in the glacier as reported in temperate and polythermal glaciers⁴⁶.

Upward vertical ice motion recorded at GNSS1 provides additional evidence of the impact of meltwater on the subglacial environment. In general, the vertical component of ice motion arises from vertical straining, sliding along an inclined bed, and basal separation due to pressurized water⁴⁷. The magnitude of the acceleration was similar at GNSS1 and GNSS2, thus significant change is unlikely in the vertical strain rates. An upward inclined bed is implausible because the vertical displacement is generally downward after the uplift in December (Fig. 3a). The absence of uplift at GNSS2 on the floating ice and the coincidence of the uplift with the acceleration both strongly suggest that the uplift was due to the separation of ice from the bed by pressurized subglacial water. We assume that the basal separation reduced friction and enhanced sliding, as is commonly observed at mountain glaciers^{12,48} and reported in the Greenland ice sheet^{49,50}.

According to the long-term meteorological record at Syowa Station, weather conditions during our field measurement were unexceptional. The sum of positive degree days (PDD) during the 2021/

22 summer period (December and January) (10.6 °C d) was 40% smaller than the mean from 1989/90 to 2025/26 (18.2 °C d) (Supplementary Fig. 3), suggesting that surface meltwater on Langhovde Glacier is generally more abundant than we observed in 2021/22. Therefore, we assume that meltwater-driven glacier acceleration is common at Langhovde Glacier and is likely to occur at other outlet glaciers in the region. Liquid precipitation has been reported seven times since December 1989 (Supplementary Fig. 3), i.e., rainfall occurs approximately every five years. From 1989 to the present, no significant trend is observed in summer mean temperature, PDD, or the frequency of liquid precipitation (Supplementary Fig. 3). Under a warming climate projected in Antarctica^{51,52}, meltwater production and its impact on glacier dynamics are expected to increase along the coast. Moreover, extreme weather events caused by atmospheric rivers will become more frequent and severe in the future⁵³. Such events trigger intense surface melt and rainfall by transporting heat and moisture from the subtropics and mid-latitudes to the Antarctic coast⁵⁴. Thus, further study is required to understand future climate and the influence of meltwater on outlet glacier dynamics.

Subglacial water pressure and hydraulic connection to the ocean

Subglacial water pressure at BH2201 and BH2202 was close to the ice overburden pressure (93–97% of flotation), a similar condition to those reported at large ice streams in West Antarctica. At Whillans Ice Stream, ~300 km from the grounding line, subglacial water pressures measured in >20 boreholes were consistently very high, typically 98–100% of the overburden pressure for ice that is ~1000 m thick⁵⁵.

Similarly, water pressure beneath the ~2150-m-thick Rutford Ice Stream, approximately 40 km from the grounding line, was 97% of the overburden pressure⁵⁶. Despite the difference in ice thickness and the distance to the grounding line, similarly high water pressures were observed under the glaciers. Such conditions are favorable for basal motion and thus facilitate the fast flow of outlet glaciers.

The semidiurnal pressure variations at BH2201 and BH2202 were correlated with ocean tides recorded at Syowa Station (Fig. 3b and Supplementary Fig. 4a–c). The agreement between the tides at Langhovde Glacier and Syowa Station was confirmed by pressure measurements in the borehole BH2203 drilled below the grounding line (Fig. 1b–d and Supplementary Fig. 4a, d, e). Therefore, the glacier base at boreholes BH2201 and BH2202 was hydraulically connected to the sub-shelf cavity (Supplementary Fig. 1). Although such a connection to the ocean is likely to occur near the grounding line, no observational evidence has ever been reported previously. Salinity was not detected in the borehole, ruling out the possibility of seawater intrusion into the subglacial hydrological system, as suggested to be the case for the grounding zones of large glaciers in West Antarctica^{57,58}. Thus, only pressure changes due to ocean tides, and not seawater itself, were transmitted to the glacier base through the subglacial drainage system. Tidal influence on subglacial pressure was speculated to account for diurnal water pressure variations in boreholes drilled at Whillans Ice Stream in West Antarctica⁵⁵. In that case, however, the pressure variations were out of phase with ocean tide, suggesting a possible influence of tidally modulated glacier flow on the subglacial hydraulics^{59,60}.

Despite the hydraulic connection, the water level in BH2201 was higher than the level in BH2203 by 9–44 m. The difference generates the hydraulic gradient above the grounding line, which is crucial information for computing glacier sliding speed in a numerical ice sheet model. This is because the rate of basal ice motion is controlled by subglacial water pressure, particularly when the pressure is close to the flotation level⁶¹. Assuming a distance from BH2201 to the grounding line of 1 km, the difference in the hydraulic heads corresponds to hydraulic gradients of $0.9\text{--}4.4 \times 10^{-2}$.

Meltwater supply to the glacier base implies water is discharged into the sub-shelf ocean cavity through the grounding line. The hydraulic connection of our boreholes to the ocean also suggests seaward subglacial drainage driven by the observed hydraulic gradient. Subglacial discharge plays a critical role in melting and circulation under ice shelves⁶². Previous oceanographic measurements under the Langhovde ice shelf showed density-driven water circulation in the cavity and the intrusion of Modified Warm Deep Water into the grounding zone³⁷. Our observations confirm that subglacial discharge and subsequent upwelling likely contribute to the ocean circulation under the ice shelf.

Sub-shelf fauna near the grounding line

Under the grounding zone of Langhovde Glacier, a thin (<10 m) seawater layer extended ~1 km upglacier beyond the estimated grounding line (Fig. 1d and Supplementary Fig. 1). Our borehole observations revealed the existence of sub-shelf fauna in the region, which provides insights into the oceanic conditions near the grounding line.

Circulation under the ice shelf transports nutrients and organic matter to the grounding zone of Langhovde Glacier, which facilitates the biological activity observed in previous borehole studies³⁶. In this study, we found sessile organisms on a boulder situated directly below the borehole drilled at BH2203 (Fig. 2d), where the water layer was only ~3 m thick beneath 474 m of ice. Benthic organisms observed by our borehole video camera include a sea anemone and several different types of stalked sponges (*Homaxinella balfourensis*) (Fig. 2e, f and Supplementary Fig. 5). Similar species have been commonly observed in the Southern Ocean^{63–65}. However, those found under Langhovde Glacier are morphologically distinct from those reported in other sub-

shelf cavities in Antarctica⁶⁶. The organisms were attached to the sides of a boulder of approximate size $1 \times 0.5 \times 0.5$ m sitting on a soft-sediment seafloor (Fig. 2d). Presumably, the steep side surface offers a favorable environment to prevent the deposition of sediment released from melting ice.

Rigid substrate sessile communities are rarely observed under Antarctic ice shelves, where mobile taxa are considered dominant⁶⁶. The hypothesis is that passive suspension feeders are unfavorable for securing food in oligotrophic environments. The inhabitation of sessile organisms under the borehole implies active transport of food from open water to the thin water layer near the grounding line. Active sub-shelf circulation likely aided the migration of *Homaxinella balfourensis*, as it expands its habitat by distributing larvae through water currents⁶³.

Our drilling revealed that the glacier base is exposed to seawater over an area broader than expected from the hydrostatic assumption. Active seawater inflow to the grounding zone suggests that the thin water layer is subject to oceanic heat transport, which likely drives sub-ice-shelf melting. As reported in Whillans Ice Stream and Thwaites Glacier in West Antarctica^{57,58}, thin seawater layers are widespread along the Antarctic ice sheet margins. Thus, further in-situ investigation is essential to better understand basal melting, ice dynamics, marine biology, and geochemistry near the grounding line.

Methods

We drilled boreholes with a hot-water drilling system consisting of three high-pressure hot-water machines (Kärcher HDS1000BE) at a mean drilling rate of approximately 40 m h^{-1} (refs. 36,67). The ice thickness was determined to an accuracy of approximately ± 1 m, using water pressure data from a CTD profiler and the borehole water level measured with a tape measure equipped with a float. A CTD profiler (IDRONAUT Ocean Seven 304) was lowered into the boreholes using a Kevlar string to measure water temperature, salinity, and depth to accuracies of 0.005°C , $7 \times 10^{-3} \text{ mS cm}^{-1}$, and 0.5 m. Sampling intervals were 1 s, which corresponded to ~0.1–0.5 m in depth. A video camera (Panasonic HX-AIH) enclosed in a pressure container was lowered into the boreholes at the same time by linking together with the CTD profiler. Video files were downloaded when the camera was retrieved from the boreholes.

Subglacial water pressure was measured at the bottom of the boreholes BH2201, BH2202, and BH2203 using water pressure sensors (Valeport miniIPS for BH2201 and Geokon 4500S for the other two boreholes). In the borehole BH2201, the sensor was installed as a part of a ploughmeter designed for sliding measurement⁶⁸. The measurement accuracies/resolutions of the sensors were $\pm 1 \text{ kPa}/0.1 \text{ kPa}$ (BH2201 and BH2203) and $\pm 7.5 \text{ kPa}/2 \text{ kPa}$ (BH2202). The pressure data were corrected for atmospheric pressure, which was recorded every hour at an automatic weather station located ~2 km downglacier of BH2201 and BH2202 (Fig. 1b). Ocean tide data are available at Syowa Station (Fig. 1a), where year-around sea-level measurements with pressure sensors have been performed since 1979 (ref. 69). We used a data set with 30 s intervals distributed by the Hydrographic and Oceanographic Department, Japan Coast Guard.

Ice speed was measured at GNSS1 and GNSS2 (Fig. 1b) with dual-frequency GNSS receivers (Enabler GEM-1 and GEM-2) from 19 December 2021 to 6 February 2022. We installed 2-m-long aluminum poles in the ice and mounted GNSS antennae at the top of the poles. The GNSS receivers continuously recorded satellite signals with a sampling frequency of 1 s. We post-processed the data with those recorded at a reference GNSS station fixed on a stable rock on a side moraine (Fig. 1b, GNSS Fix). GNSS processing software RTK-LIB (<https://www.rtklib.com/rtklib.htm>) was used to perform static differential carrier-phase positioning with a broadcast ephemeris to obtain three-dimensional coordinates of the stake positions every 15 min with an accuracy of 3–5 mm (ref. 70). We computed horizontal ice speed after smoothing the time series of x -, y - and z -coordinate data with a Gaussian filter having a

bandwidth of 1 h (Supplementary Fig. 6). This bandwidth was chosen so that the mean deviation of the data from the smoothed curve was equivalent to the positioning accuracy. The relative error in the computed hourly ice speed was 7–12% for the mean speed at GNSS1, and the actual error after the smoothing procedure was smaller than this estimate. Data from GNSS1 was unavailable for the periods of 24–25 December 2012, 12–19, and 22–26 January 2022 because of device failure. Ice speed during the period was computed by estimating the coordinates with spline interpretation (Supplementary Fig. 6).

Ice temperature was measured with thermistor strings (Geokon Model 3810) installed in the boreholes drilled in 2011/12 (BH1202 and BH1204) and 2017/18 (BH1804) (Fig. 1b–d). Thermistor sensors were placed at 50, 100, 150, 200, 250, 300, 325, 350, 375 and 400 m below the glacier surface in BH1202; 26, 76, 126, 176, 226, 276, 326, 376 and 426 m in BH1204; and 30, 130, 180, 205 and 230 m in BH1804 (Supplementary Fig. 2). The accuracy of the thermistor sensor, as guaranteed by the manufacturer, was $\pm 0.2^\circ\text{C}$. However, our measurement accuracy improved by calibrating each sensor in ice water and correcting for the cable resistance. Temperature measurements were taken approximately one year after installation, when the site was revisited the following season.

Aerial photographs were taken over the drilling sites BH2021 and BH2022, covering an area of approximately 500×500 m, by operating a drone (DJI Phantom 4 Pro V2) 100 m above the surface. We repeated the survey on 1, 3, and 13 January 2022 to generate orthorectified mosaics with the software Agisoft Metashape.

During the field campaign, a basic automatic weather station (Vaisala WXT530) was operated on the glacier at 54 m a.s.l. (Fig. 1b). The sensor recording air temperature and liquid precipitation every 5 min was installed 2 m above the ice surface and connected to a data logger (Campbell Scientific, CR1000X). Daily mean temperature and daily cumulative precipitation were calculated for plotting the data. Daily mean, minimum, maximum temperatures, and a rainfall record at Syowa Station are distributed by Japan Meteorological Agency (<https://www.data.jma.go.jp/antartic/data/report/index-e.html>).

The flotation level z_f in Fig. 1d was computed as

$$z_f = z_b + (z_s - z_b) \frac{\rho_i}{\rho_s}, \quad (1)$$

where z_b and z_s are ice base and surface elevations, and ρ_i ($=910 \text{ kg m}^{-3}$) and ρ_s ($=1025 \text{ kg m}^{-3}$) are ice and seawater densities. This is the expected elevation of the water surface for floating ice in the hydrostatic equilibrium condition.

Data availability

The Landsat images are distributed by USGS at <http://earthexplorer.usgs.gov/>. The Pléiades digital elevation model was provided by the Pléiades Glacier observatory initiative of the French Space Agency and Laboratoire d'Etudes en Géophysique et Océanographie Spatiales. The data presented in the paper are deposited and available at Mendeley Data⁷¹.

Code availability

The code used for plotting the figures is deposited and available at Mendeley Data⁷¹.

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Acknowledgements

We thank the 63rd Japanese Antarctic Research Expedition (JARE63) members for their support of the field activity on Langhovde Glacier. C. Richter, W. Hagen, H. Numanami, and Y. Watanuki helped with identification of the organisms observed under the ice shelf. Drilling equipment was constructed by the machine shop at the Institute of Low Temperature Science, Hokkaido University. This research was performed as a scientific program of JARE63 as Prioritized Research Project. S.S. discloses support for the research and publication of this work from JSPS KAKENHI Grant Numbers 20H00186, 24H02339 and 25H00452.

Author contributions

S.S. designed the research and wrote the manuscript. S.S., K.K., M.M. and A.W. drilled boreholes. S.S., K.K. and M.M. performed field measurements and data analyses. S.S., K.K., M.M. and A.W. discussed the results and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41467-026-72724-x>.

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Peer review information *Nature Communications* thanks Douglas Benn, Samuel Doyle and Ian Willis for their contribution to the peer review of this work. A peer review file is available.

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